Muhammad Noman Ashraf

M.Sc. Electrical Engineer

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Personal Details

Date of Birth: 13, Sep 1993

Language: Urdu, English and Korean

Address: AbuDhabi,UAE

RESEARCH INTERESTS

1. Multilevel Single/ Three Phase Inverter **2.** Isolated/Non-Isolated DC-DC Converters **3.** Hybrid AC/DC Grid **4.** V2G and G2V

5. High frequency Resonant converter for Electric Vehicles application

EDUCATION

PhD in Engineering	Khalifa University AbuDhabi, UAE	CGPA: 3.68/4.0	2022- Present
M.Sc. Electrical Engineering (Power Electronics)	Soongsil University Seoul, South Korea	CGPA: 4.39/4.5	2017-2019
B. Sc. Electrical Engineering (Electronics)	Pakistan Airforce Karachi Institute of Science and Technology (PAF-KIET) Karachi, Pakistan	CGPA: 3.7/4.0	2012-2016

MASTER THESIS

- **Digital Lock-In Amplifier Based Harmonic Compensation and Islanding Detection for Single Phase Grid Connected Inverter Systems**
  1. Less than 5% THD achievement 5kW Bi-directional Inverter for complete Load Range under distorted grid
  2. Design of Simple and Robust Islanding Detection Algorithm for single phase systems.
  3. Complete MPPT based PV Grid Connected system with DC Link voltage stabilization equipped with Islanding Detection.

WORK EXPERIENCE

1. Associate Researcher, OKY LTD South Korea

Research Department
OKY LTD
Seoul, South Korea

From 2019- Present

1. PROJECT: Complete Design and testing 10kW Dual Stage Bi-directional DC-DC Converter.

Responsibilities:

- The topology and design verification in PSIM.
- Designing of Circuit, Inductors, Transformers, PCB layout and complete circuitry.
- Each power component selection for prototyping.
- Complete hardware and software Implementation with TI Digital Signal Processor (DSP).
- Complete system experimental verification with bidirectional power flow.

2. Single Phase Inverter based Dynamic Voltage Compensator (DVC) Software and Algorithm Design

- Complete Algorithm design for single phase inverter for voltage SAG compensation.
- Design of Supercapacitor charging algorithm.
- Implementation of Fast sag detection and compensation with Inverter
- Smooth transition of load from grid to inverter within 2mS.

3. Technical Manager of all single-phase products

- Firmware changes history management of all models
- Hardware changes version management of all models.
- Supervise the Quality Control Experiments.

4. New Projects Proposal Writing

- Topology analysis and simulations for the new project proposals.
- Efficiency and loss estimation.
- Peak ratings estimation for physical components.
- Approximate cost estimation with major component selection.

2. RESEARCH ASSISTANT at Soongsil University

Renewable Energy Conversion Laboratory
Soongsil University
Seoul, South Korea

2017-2019

5. Complete Design and testing of 5kW Bi-directional Grid Connected Inverter System for AC/DC Micro grid.

- Funded by South Korean Electric Corporation.
- The design verification by PSIM and MATLAB software.
- Designing of Circuit, Inductors, PCB layout and complete circuitry.
- Controller design for bi-directional power flow. Grid Connected Mode and Rectification Mode
- Complete hardware and software Implementation with Digital Signal Processor (DSP) **TMS320F28335**.

6. Two Stage Photovoltaic DC/AC Conversion System (Dual Active Bridge as DC-DC converter and Single-phase inverter as DC/AC converter);

- Implementation of MPPT Algorithm (Ripple Correlation Method) for two stage Grid Connected system.
- Average Voltage Control at the DC-Link.
- Robust Grid Current Control by using Rotatory Reference frame.
- Improve Harmonic Compensation of inverter to make THD within 5% to satisfied IEEE 1547 and 517 harmonic standards.

7. Inverter Mode change in Quarter Line Cycle and DC Link Voltage Stabilization

- Mode change from grid connected to rectification within Quarter Line Cycle.
- DC link voltage stabilization in case of power deficiency in DC Grid side.
- Synchronous Rectification to inject power from grid to DC Bus.

8. Current Controller for Grid Connected Mode and Rectification Mode

- Synchronous and Stationary Reference Frame Controller Designing for Current Control.
- Complete designing done in SISO TOOL MATLAB.
- Type II Compensator for Rectification Mode, reverse power flow.

9. Novel Harmonic Compensation Method Design for Distorted Grid Condition

- Complete Algorithm Design to achieve less than 5% THD for Inverter.
- Less than 5% THD for all rated power. i.e., even low power.
- Ensuring all the rules IEEE 1547 and 517 harmonic standards.

10. Development of Novel Islanding Detection Method for Grid Connected Inverter

- Designing of non-invasive sensor less Islanding Detection method for Inverters
- Disconnection from Grid within 5-Line cycles in case of Fault.
- Cost effective, simple and robust. Implementation tested under real grid with 5kW Inverter.

11. Complete Design of Digital to Analog Converter (DAC) SPI Serial Interface for Inverter

- Complete Designing of the DAC Circuit for Inverter.
- Real time DSP Interface with DSP through SPI Serial.
- Up to 50Mhz Baud rate serial communication with Digital Signal Processor (DSP).

3. Lab Engineer

PAF-KIET
Main Campus.
Karachi, Pakistan.

Jan-2017
Sep-2017

1. Instrumentation and Measurement Teaching Assistant.

- Lab Instructor of Instrumentation & Measurement Laboratory to undergrad students.
- Performing practical, designing and simulation of aircraft models in LabVIEW and MATLAB Simulations.
- Sensors interfacing with hardware circuits and debugging.

4. Design Engineer

TECHNIA-3M
Karachi, Pakistan.

July- 2016
Dec- 2016

1. **Design Engineer for Designing and debugging of Quadcopters**

- Worked on SITL and HIL of Pixhawk Px4 Autopilot controller.
- Hardware in Loop simulations and debugging.

5. Research Assistant

NED University of Science and Technology
Main Campus.
Karachi, Pakistan.

June 2016-
Aug 2016

- Worked Under the supervision of Pakistan's renowned Scientist Dr. Khurram of CS Department at CIS Robotics Lab
- We equipped the simple 3D prosthetic hand with the vibrotactile feedback for disabled People.
- My specific task was to program a controller and interface sensors placed on prosthetic with it.
- Our research work was published in IEEE ICRAI 2016 and it was also accepted at **ASIA HAPTICS JAPAN** 2016 for publication.

P U B L I C A T I O N S

	1. M. N. Ashraf, M. S. E. Moursi, K. Al Hosani and A. Al Durra, "Andronov–Hopf Oscillator-Based Phase-Locked 2. Loop for Three-Phase Grid Connected Inverters," in IEEE Transactions on Power Electronics, vol. 40, no. 6, pp. 3. 7634-7639, June 2025, doi: 10.1109/TPEL.2025.3531901. (Published)
	4. M. N. Ashraf, M. S. E. Moursi, K. Al Hosani and A. Al Durra "Andronov-Hopf Oscillator Based Harmonic 5. Elimination Scheme for Grid-Connected Inverters" IEEE Transactions on Smart Grid, (under review)
Journal	6. M. N. Ashraf, R. A. Khan and W. Choi, "A Novel Selective Harmonic Compensation Method for Single-Phase Grid-Connected Inverters," in IEEE Transactions on Industrial Electronics, vol. 68, no. 6, pp. 4848-4858, June 2021, doi: 10.1109/TIE.2020.2989723. (Published)
	7. Khan, R.A.; Ashraf, M.N.; Choi, W. A Harmonic Compensation Method Using a Lock-In Amplifier under Non-Sinusoidal Grid Conditions for Single Phase Grid Connected Inverters. Energies 2021, 14, 597. https://doi.org/10.3390/en14030597 (Published)
	8. Amin, S., Ashraf, MN, & Choi, W. (2020). A new harmonic compensation method based on digital lock-in amplifiers for grid-connected inverters. Journal of Power Electronics Society, 25 (5), 358–368. https://doi.org/10.6113/TKPE.2020.25.5.358 (Published)

	9. Muhammad Noman Ashraf and Woojin Choi, "A Novel Hybrid Islanding Detection Method Using Digital Lock-In Amplifier", in Annual Summer Conference of the Korean Institute of Power Electronics (KIPE) , South Korea, Jun. 2019 (Published).
	10. Muhammad Noman Ashraf, Reyyan Ahmed Khan and Woojin Choi, "A Novel Harmonic Compensation Technique for the Grid-Connected Inverters", in Annual Summer Conference of the Korean Institute of Power Electronics (KIPE), South Korea, Jun. 2019 (Published).
	11. Muhammad Noman Ashraf, Reyyan Ahmad Khan and Woojin Choi, "A Robust PLL Technique using a Digital Lock-In Amplifier under the Non-Sinusoidal Grid Conditions" in International Conference on Power Electronic (ICPE 2019), IEEE, Busan, South Korea, May 27-30, 2019 (Published).
	12. Muhammad Noman Ashraf, Reyyan Ahmed Khan and Woojin Choi, "A Robust PLL Technique using a Digital Lock-In Amplifier under the Non-Sinusoidal Grid Conditions", in Annual Winter Conference of the Korean Institute of Power Electronics (KIPE), South Korea, Nov. 2018 (Published).
	13. Reyyan Ahmad Khan, Muhammad Noman Ashraf and Woojin Choi, "A Robust Harmonic Compensation Technique using Digital Lock-in Amplifier Under the Non-Sinusoidal Grid Voltage Conditions for the Single Phase Grid Connected Inverters" in International Conference on Power Electronic (ICPE 2019), IEEE, Busan, South Korea, May 27-30 , 2019 (Published).
Conference Papers	14. Reyyan Ahmad Khan, Muhammad Noman Ashraf and Woojin Choi, "A Robust Harmonic Compensation Technique for thr Single Phase Grid Connected Inverters Under the Non-Sinusoidal Grid Voltage Conditions", in Annual Winter Conference of the Korean Institute of Power Electronics (KIPE), Seoul, South Korea, Nov. 2018 (Published).
	15. Reyyan Ahmad Khan, Muhammad Noman Ashraf and Woojin Choi, "A Novel Harmonic Compensation Method for the Single-Phase Grid Connected Inverters", in Annual Summer Conference of the Korean Institute of Power Electronics (KIPE), Kangwon province, Gangwon, South Korea, July 3-6, 2018 (Published with Best Conference Paper Award).
	16. Reyyan Ahmad Khan, Muhammad Noman Ashraf and Woojin Choi, "A Method to Improve the Performance of Phase-Locked Loop (PLL) for a Single-Phase Inverter Under the Non-Sinusoidal Grid Voltage Conditions", in Annual Winter Conference of the Korean Institute of Power Electronics (KIPE), Seoul, South Korea, Nov. 2017 (Published with Best Conference Paper Award).
	17. Saghir Amin, Muhammad Noman Ashraf and Woojin Choi, "Power Decoupling Control of the Two-Stage Single-Phase DC-AC Power Conversion System to Eliminate the Double Line Frequency Ripple", in Annual Summer Conference of the Korean Institute of Power Electronics (KIPE) , South Korea, Jun. 2019 (Published).

	18. Kashan Aqeel, Muhammad Noman Ashraf, Mudassir Ibrahim Awan and Nabeel Ahmed (2016, November). Vibrotactile feedback for 3D printed prosthetic Hand: Presented at "IEEE 2nd International Conference on Robotics and Artificial Intelligence (ICRAI 2016) ", NUST, Islamabad. (Published)

TECHNICAL SKILLS

- Practical Power Electronics with Hands On experience.
- Good knowledge & work experience in TI DSP from Texas Instruments (TI) and PIC Microchip Technology.
- Good knowledge in Embedded System Programming (C/C++)
- Circuit, Control Design and PCB Design(ORCAD) for Power and Logic Circuit
- Experience in Single phase inverter modelling and control design.
- Experience as Team leader and also working as team member.
- Ability to handle and prioritize multiple assignments and Fulfil deadlines.
- Software Skills: PSIM, MATLAB, Orcad, PSPICE, AutoCAD, Proteus, PCB Express , C++, Assembly Language, MS Office

AWARDS AND ACHEIVEMENTS

- International Graduate Research Scholarship from 2017 to Jul. 2019 (Awarded by **Soongsil University**, Seoul, Korea)
- University Merit Scholarship for Engineering Student (4 years Awarded by PAF-KIET, Pakistan)
- In Top 5 among all Engineering Student in throughout 4 year.
- Research paper acceptance at ASIA HAPTICS **Japan 2016**.
- Got 1st positions in declamation competitions at PAF-KIET.
- Dean's Honours Badge in 1st ,3rd, and 6th semester.
- President's Honours Badge in 2nd and 7th semester
- College Honours Badge in 4th, 5th and 8th semester at PAF-KIET.

REFERENCES

Professor Mohamed El Moursi Center Director ,Advanced Power and Energy Center Khalifa University, AbuDhabi, UAE mohamed.elmoursi@ku.ac.ae	**Professor Ashraf Ahmad, Ph.D.** Soongsil University, Seoul, South Korea Department of Electrical Engineering Email: ahmed@ssu.ac.kr Mobile: +82-10-2154-3282